Devops – Exercise 1

```
PS D:\vivek\7th_Sem\devops\excercise1> choco install minikube
Chocolatey v2.7.4
3 validations performed. 2 success(es), 1 warning(s), and 0 error(s).

Validation Warnings:
 - A pending system reboot request has been detected, however, this is
   being ignored due to the current Chocolatey configuration.  If you
   want to halt when this occurs, then either set the global feature
   using:
     choco feature enable --name="exitOnRebootDetected"
   or pass the option --exit-when-reboot-detected.

Installing the following packages:
minikube
By installing, you accept licenses for the packages.
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading kubernetes-cli 1.37.1... 100%

kubernetes-cli v1.37.1 [Approved]
kubernetes-cli package files install completed. Performing other installation steps.
The package kubernetes-cli wants to run 'chocolateyInstall.ps1'.
Note: If you don't run this script, the installation will fail.
Note: To confirm automatically next time, use '-y' or consider:
choco feature enable -n allowGlobalConfirmation
Do you want to run the script?([Y]es/[A]ll scripts/[N]o/[P]rint): Y

Extracting 64-bit C:\ProgramData\chocolatey\lib\kubernetes-cli\tools\kubernetes-client-windows-amd64.tar.gz to C:\ProgramData\chocolatey\lib\kubernetes-cli\tools...
C:\ProgramData\chocolatey\lib\kubernetes-cli\tools
Extracting 64-bit C:\ProgramData\chocolatey\lib\kubernetes-cli\tools\kubernetes-client-windows-amd64.tar to C:\ProgramData\chocolatey\lib\kubernetes-cli\tools...
C:\ProgramData\chocolatey\lib\kubernetes-cli\tools
 ShimGen has successfully created a shim for kubectl-convert.exe
 ShimGen has successfully created a shim for kubectl.exe
 The install of kubernetes-cli was successful.
  Deployed to 'C:\ProgramData\chocolatey\lib\kubernetes-cli\tools'
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading Minikube 1.39.0... 100%

Minikube v1.39.0 [Approved]
Minikube package files install completed. Performing other installation steps.
 ShimGen has successfully created a shim for minikube.exe
 The install of Minikube was successful.
  Deployed to 'C:\ProgramData\chocolatey\lib\Minikube'

Chocolatey installed 2/2 packages.
 See the log for details (C:\ProgramData\chocolatey\logs\chocolatey.log).
```

```
PS D:\vivek\7th_Sem\devops\excercise1> minikube start --driver=docker
* minikube v1.39.0 on Microsoft Windows 11 Home Single Language 26H2
* Using the docker driver based on user configuration
* Using Docker Desktop driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.51 ...
    > gcr.io/k8s-minikube/kicbase:  507.61 MiB / 507.61 MiB  100.00% 6.93 MiB p
* Creating docker container (CPUs=2, Memory=4000MB) ...
* Preparing Kubernetes v1.37.0 on containerd 2.3.4 ...
* Configuring CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, default-storageclass

! D:\Docker\Docker\resources\bin\kubectl.exe is version 1.32.2, which may have incompatibilities with Kubernetes 1.37.0.
  - Want kubectl v1.37.0? Try 'minikube kubectl -- get pods -A'
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
PS D:\vivek\7th_Sem\devops\excercise1> kubectl run hello-k8s --image=nginx --port=80
pod/hello-k8s created
PS D:\vivek\7th_Sem\devops\excercise1> kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
hello-k8s   1/1     Running   0          22s
PS D:\vivek\7th_Sem\devops\excercise1> kubectl expose pod hello-k8s --type=NodePort --port=80
service/hello-k8s exposed
PS D:\vivek\7th_Sem\devops\excercise1> minikube service hello-k8s
```

| NAMESPACE | NAME | TARGET PORT | URL |
|---|---|---|---|
| default | hello-k8s | 80 | http://192.168.49.2:30220 |

```
* Starting tunnel for service hello-k8s.
```

| NAMESPACE | NAME | TARGET PORT | URL |
|---|---|---|---|
| default | hello-k8s | | http://127.0.0.1:58163 |

```
* Opening service default/hello-k8s in default browser...
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```

# Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

*Thank you for using nginx.*